

Universe24: the group of order 24 behind the clicks, one local integer operator on the cube’s lattice, and the quantum and relativistic experiments it confronts

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Abstract

Can one local integer operator on a cubic lattice, read only through detectors’ clicks, confront the quantum and relativistic experiments? The whole GameBoard is one element of one integer module, and the law is one linear operator on it: at every Node a second-order step on the Node’s own two levels and its six neighbours, with one declared integer pair per family and the remainder of the one division kept, and at declared cells a coupling between two families and an integer label matrix. The lattice’s symmetries are the 48 signed permutations of the axes and their 24 rotations. Two acts complete the law. The click is the only non-local act: when a record’s one-way inward flux into a detector, a cube of side at least 3, crosses the record’s threshold, the record ends whole, everywhere, at once. The birth is the only write, the other side of a click. Exact on the lattice: the conserved quadratic form of every family and of two coupled families, the step’s bijection, the local flux identity, one click per quantum, Born’s rule as a detector’s share of a record’s inward flux to the grain $1/W$ of the record’s wheel, and no signalling through the click. A product state gives the Clauser–Horne–Shimony–Holt (CHSH) sum at most 2; an entangled pair read on integer polariser axes gives $14/5$. Recovered as the wave number goes to 0: $c = 1/\sqrt{3}$ as the band’s pace, isotropic at second order by the 48; and a moving packet’s clock slows by its own dispersion, with no rule for it. An experiment is a list of entries on the board; its initial state is generated once by the operator itself and checked in integers at load. Seventeen experiments are closed in the algebra with pins written blind; none is run yet on this engine.

Keywords: integer lattice model; the symmetry group of the cube; one linear operator; entangled pair; Bell correlations; discrete relativity; pre-registered simulation

1 Introduction

The group was found, not assumed. The operations came first, as acts on the board, and the road is told in Section 7: a message crossed one Link per interval, messages met at a Node, a detector clicked. The group is the answer to which maps preserve what the acts work on, a Node with six Ports and its Links and a causal cone of one Link per interval: the bijections of the six Ports that keep opposite Ports opposite are the 48 signed permutations of the axes, and the only $\mathbb{Z}$ -linear bijections that preserve the cone are these 48, so no boost is among them and Lorentz’s form is reached Outside from the clicks, not as a symmetry of the lattice. The 24 came from the hand, a pseudoscalar that a reflection flips and a rotation keeps, which splits the 48 into the 24

rotations and the 24 reflections. Every rule is required to commute with the 48, a requirement beyond locality. Today the law is one operator on one element, with no stepping body and no merge of messages [9, 9.19].

The whole move, in the order it is built. Seven things, each named once in the project's terminology [32].

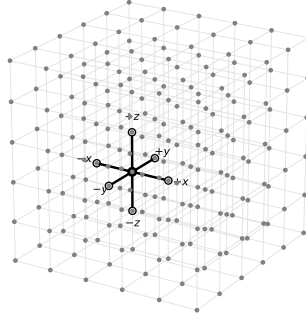
- (1) *The GameBoard and its group.* The board is a cubic lattice of Nodes, six neighbours each. The symmetry group of the cube is the group of order 48: the 24 rotations, the group of order 24, and the reflections. The law is the same at every Node and commutes with the 48 and with the translations, and two things follow: the pace of light is the same in every direction at leading order, $c^2 = 1/3$ from the six reads, and the departure from it appears only in a short wave, in the fixed cubic pattern of the 48 (Eq. (11)).
- (2) *The one element.* The state of the whole board, every Node, every family and every label together, is one element $\mathbf{v}$ of one integer module: one summand per quantum, a *record*, with its two levels (now and one interval before) at every Node and label and the remainder of the one division [9, 9.19 (1)]. A *family* is an index of the summands: its vacuum pair [num, den] and its label module, and nothing else [9, 9.22 (1)]. A world has at most three: light, the experiment's matter and the holder of the tools' bodies.
- (3) *The one operator.* The law is one linear map $\mathbf{S}$ of that element (Eq. (1)): per family the second-order step with the family's pair at every Node, and at declared cells a coupling between two families and an integer label matrix [9, 9.19 (2)].
- (4) *The tools.* A tool is a region of the operator: a cube of the holder family with its material integers (a pair, a coupling, an axis), its total momentum $\mathbf{K}$ (the wave vector of its character, $\mathbf{0}$ at rest), at most a birth, and a name [9, 9.19 (5)]; no code belongs to a tool [10, A].
- (5) *The detector and the click.* A detector is a named region, a cube of side at least 3. The click is the only non-local act: a record's one-way inward flux into the detectors named on its ladder, summed over intervals, crosses the record's threshold, and the record ends whole, everywhere, at that interval; the click is the detector's and names no Node inside it [9, 9.25].
- (6) *The birth.* The only write, the other side of a click: the click of an emitter body's own excited record writes the born record's two levels once on the emitter's cells, the transpose $\mathbf{E}^T$ of the reading, and from then on only the law advances it [9, 9.13, 9.17].
- (7) *The experiment.* A list of entries on the board, each its family, how much and where, with its momentum; a generator writes the initial state once from the operator itself, the engine checks it in integers at load and runs the law, and the output is the clicks, compared with a pin written before the run (Section 2, "The experiment as data"; Section 6).

The general formula. The law is one step $\mathbf{S}$ of the one element at every interval, and two acts on its summands:

the step: $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M}_F \mathbf{a}_{\text{now}} + \mathbf{c}_F$, $\mathbf{M}_F = \mathbf{D}_F^{-1} \mathbf{A}/3$, per family F, with the remainder kept; the click: $2W C(t^*) \geq (2u + 1) T$, $C(t) = \sum_{s \leq t} \sum_{\text{Ports } j \rightarrow i \text{ into the ladder's detectors}} \max(G_{ij}(s), 0)$; the birth: $(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}}) \leftarrow \mathbf{E}^T(\text{the born pair})$ on the emitter's cells, at the click's interval.

(1)

In the step, $\mathbf{a}_{\text{now}}$, $\mathbf{a}_{\text{before}}$ and $\mathbf{a}_{\text{next}}$ are a record's levels over the Nodes now, one interval before and one interval on; $\mathbf{A}$ is the read matrix of the board (the six reads of every Node); $\mathbf{D}_F$ is the diagonal matrix of the per-Node weights $\text{den}_i/\text{num}_i$ of the family's pair; and $\mathbf{c}_F$ is the coupling's first difference of the other family's levels at the cells of a coupled body, with, at the cells of a polar



one Node and its six Ports, the Links to its six neighbours

Fig. 1 The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x, \pm y, \pm z$ (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it, whose convex hull is the octahedron of Figure 2. Each record keeps at a Node its two levels and its remainder there, and a body's region keeps its material integers; nothing else sits at a Node. Drawn from the definitions; no run.

body, the integer label matrix on the record's two label rows. In integers the step is Eq. (2), one division by 3 den_i with the remainder kept. In the click, G_{ij} is the flux into Node i from Node j (Eq. (5)), T the record's conserved form, u its residue and W its wheel, both read from the law at its birth (Definition 1); the click falls at the first interval t^* at which the inequality holds, at the detector whose share of that interval's flux holds the threshold, and deletes the record. The birth is fired only by a click. Nothing else happens on the board: no table, no rate, no drive, no take and no declared residue. Section 2 states each piece with its proof or its mark; every derivation of the paper names the piece it starts from.

Physics as the passage of information. What the law describes is the passage of information from an emitter to a detector, and nothing between them is read. A record is born at an emitter's click with its two levels written once; it moves on the lattice by the one operator, at most one Link per interval; it leaves at a click, where the record ends whole, and the click is what is read: reality Outside is the clicks, and a number of the board is a diagnostic. Transport is local at every step; the one non-local act is the click itself, the record ending everywhere at once, and no information passes through it (Theorem 4). The road by which this object was found is Section 7; the engine on which it is computed is Section 6.

The reading rule and the notation. Only a detector's click is a measurement, and an experiment is compared with nature and with its pin through its receivers' clicks alone; the operator, the tools, conservation and the initial state are proved in the algebra and checked by the property test and the load check, never by an experiment's run [9, 9.21 (4)]. Every physics name in this paper (Born's, Tsirelson's, Malus's, Lorentz's) names the known form that the law's result recovers. A number of the GameBoard's state is a diagnostic of the host, labelled so, never compared with nature and never pinned. Every symbol is named in English at its first use, a scalar plain, a vector in bold lowercase, a matrix or an operator in bold uppercase.

What this paper contributes, and what it does not claim. New here: one explicitly stated integer operator with its click and its birth, whose conserved forms, bijection, flux identity, Born's rule and no-signalling are theorems on the lattice; experiments as short lists of entries, their initial states generated from the operator itself and checked in integers at load; and seventeen experiments

closed in the algebra with pins written blind before any run. Not claimed: a derivation of the Lorentz group, which is not among the 48; a moving rigid rod, which the model does not yet define (it has clocks and no rods, [9, 9.24 (6)]); or any comparison with nature before the experiments are run on this engine. Gravity is not in this paper.

2 The algebra: one element, one operator, the click and the birth

This section states the object and nothing else: the board and its group, the one element, the one operator with its conserved forms, the flux, the click, the birth, the families and the experiment as data. Each statement carries the mark of the algebra document [9]: PROVED (a proof for every board, in the document or here), DERIVED (from proved statements, by argument), COMPUTED (a computation, not a proof), or CARRIED (a design's proof, cited).

The definitions, in order. Five objects, each one line of mathematics before its words.

- (1) The GameBoard: the Nodes of a box in $\mathbb{Z}^3$ whose axes are periodic or open; the six Ports of a Node, $P = \{\pm x, \pm y, \pm z\}$, with the involution $e \mapsto -e$ (the opposite Port); at most one Link per interval.
- (2) The group of order 48 and its rotations: $G_{48} = \{g : P \rightarrow P \text{ a bijection with } g(-e) = -g(e)\}$, the 48 signed permutations of the three axes; $\det : G_{48} \rightarrow \{+1, -1\}$ a homomorphism; $G_{24} = \ker \det$, the rotations, the group of order 24 (Theorem 1). The hand, a pseudoscalar, $h \mapsto \det(g) h$, tells the two cosets apart.
- (3) The one element: $V = \bigoplus_q V_q$, $V_q = (\mathbb{Z}^{\text{Nodes}} \otimes L_q)^2 \otimes (\text{the remainders})$, one summand per quantum q , L_q its label module ($\mathbb{Z}^2$ per arm; $\mathbb{Z}^2 \otimes \mathbb{Z}^2$ for a record of rank 2); the state of the world at an interval is one element $\mathbf{v} \in V$ [9, 9.19 (1)].
- (4) The one operator: $\mathbf{S} : V \rightarrow V$, per summand of family F the step of Eq. (1), plus the coupling between summands of coupled families and the label matrix on the axes' cells; linear, local and equivariant [9, 9.19 (2), (6)].
- (5) The two acts: the click, a record's one-way inward flux $\mathbf{E}$ against its form, the rung and the deletion of the summand, the only non-local act; and the birth $\mathbf{E}^\top$, the only write, at the click's interval (Definitions 1 and 2).

The words of this paper. The *GameBoard* is the cubic array of Nodes, each joined to six neighbours by Links through directional Ports; time is a sequence of intervals. *Inside* is the board and what moves on it, where no one measures; *Outside* is the game above the board, the detectors and their clicks, the only place a number is read. A *record* is one summand of the element, one quantum: its two levels over the Nodes and its labels, with the remainder kept per Node on the record's own row. A *family* is an index of the summands, its vacuum pair and its label module. A *body* is a region of the operator where a family's pair differs from the vacuum's; a *tool* is a region with its integers, at most a birth and a name; a *face* is the operator's zero row beyond an open border; a *receiver* or *detector* is a name the host gives a region, on records' ladders [9, 9.19 (5)]. Every number carries its *kind*: DETECTOR, a click or a count of clicks, the only measurement; GAMEBOARD, a number of the board's state, a diagnostic and never a result; COMPUTATION, a number computed from the algebra before any run; DECLARATION, a declared integer of an experiment; HOST, a number of the host's bookkeeping, never a reading. Figure 1 draws the board in space.

The six Ports, the group and the front. Everything geometric in the law follows from one Node and its six Ports, $\pm x, \pm y, \pm z$, and from at most one Link per interval. The maps of the six

Ports onto themselves that keep opposite Ports opposite are the signed permutations of the three axes, 48 in all, the symmetry group of the cube and of the octahedron. The determinant splits them into two halves of 24: the rotations, which permute the cube's four body diagonals, and the improper maps, the reflections with the central inversion among them.

Theorem 1 (the 24 of the name). *The maps of the six Ports onto themselves that send opposite Ports to opposite Ports form the group of the signed permutations of the three axes, of order $2^3 \cdot 3! = 48$. The determinant is a homomorphism of this group onto $\{+1, -1\}$; its kernel, the group of order 24 (the rotation group of the cube), has index 2 and is isomorphic to the symmetric group S_4 of the permutations of the cube's four body diagonals; the other coset, the reflections, has 24 elements too. The hand, a pseudoscalar that a map g sends to $\det(g)$ times itself, tells the 24 rotations from the 24 reflections, as does an axial vector.*

Proof. A map that keeps opposite Ports opposite is fixed by the images of $+x, +y, +z$, which must be three Ports no two of them equal or opposite: $6 \cdot 4 \cdot 2 = 48$ choices, each a signed permutation matrix, and every signed permutation matrix keeps opposite Ports opposite. The determinant of a product is the product of the determinants, so $\det$ is a homomorphism, onto since -1 has determinant -1 ; its kernel has index 2, order 24. A rotation permutes the four lines through opposite vertices of the cube, and only the identity fixes all four, so the kernel embeds in S_4 , of order 24: the two are isomorphic. $\square$

The rule per family. At every Node i a record's row $(a_{\text{now}}, a_{\text{before}}, r)$ is mapped once per interval by

$$3 \text{den}_i a_{\text{next}} + r' = \text{num}_i S_6 - 3 \text{den}_i a_{\text{before}} + r, \quad 0 \leq r' < 3 \text{den}_i, \quad (2)$$

then $a_{\text{before}} \leftarrow a_{\text{now}}, a_{\text{now}} \leftarrow a_{\text{next}}, r \leftarrow r' [9, 8.1]$. Here S_6 is the sum of the six reads of the record's a_{now} (the neighbour across a Link; the Node itself on a folded axis of extent 1; 0 beyond an open face), $[\text{num}_i, \text{den}_i]$ the family's pair at the Node (the vacuum's pair everywhere, a body's pair on its cells) and r the remainder of the division, kept on the record's row. Light is the pair $[1, 1]$; a massive family is a pair with $\text{den} > \text{num}$, and nothing else names the mass. The step is made of three verbs on the row: the bilinear form (B), one entry num_i on the six reads and one entry -3den_i on the row's past; the translation (T) of the row by that rate; and the division (D) by the wall 3den_i with the remainder kept. The three tests hold [9, 8.1]: generic, one primitive with one declared pair and no family name; vector, the three verbs with no root and no float; local, the record's own row and its six reads, nothing kept at a Node beyond the row and its remainder.

The conserved form. Let $\mathbf{A}$ be the read matrix of the board (A_{ij} the number of reads of j among the six reads of i), symmetric with row sums at most 6, $\mathbf{D} = \text{diag}(D_i)$ with $D_i = \text{den}_i / \text{num}_i$, and $\mathbf{M} = \mathbf{D}^{-1} \mathbf{A} / 3$. Without the remainder the rule is $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$, and the form

$$I(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = \sum_i D_i (a_{\text{next},i}^2 + a_{\text{now},i}^2) - \frac{1}{3} \sum_i a_{\text{next},i} (\mathbf{A} \mathbf{a}_{\text{now}})_i \quad (3)$$

is conserved on every board and every extent, since $\mathbf{DM} = \mathbf{A} / 3$ is symmetric; it is positive definite for every massive pair and semidefinite at light's pair; and with the remainders kept the change per interval is exactly $I(t) - I(t-1) = \sum_i (a_{\text{next},i} - a_{\text{before},i})(r_i - r'_i) / (3 \text{num}_i)$, an integer identity with no tolerance (PROVED [9, 8.2]). A static level has $I = 0$.

The modes and the stability condition. Restricted to one family, $\mathbf{S}$ is a second-order linear recurrence with the symmetric operator $\mathbf{D}^{-1/2}(\mathbf{A}/3)\mathbf{D}^{-1/2}$, whose eigenvectors are the board's *modes* and whose eigenvalues are $2 \cos \omega$, the angle ω being the mode's clock, its frequency per

interval; on a torus there is one mode per Node. A mode is *bound* when $2\cos\omega$ lies above the vacuum band's top and *free* inside the band. The form I is positive definite, and every mode a rotation, exactly when the largest eigenvalue is below 2; by Perron–Frobenius that one condition bounds both ends of the spectrum. A pair with $\text{num} \leq \text{den}$ at every Node is sufficient, not necessary. The loader's check is the spectral one: the composed operator's largest eigenvalue below 2, the file refused otherwise (PROVED [9, 9.19 (2)]).

The coupling of two families. At a cell of a coupled body R the two records' rows are coupled by one entry of verb (B) over the other record's two levels, the first difference both ways: the massive row gains $g(a_{l,\text{now}} - a_{l,\text{before}})$, the receive, and light's row gains $-G(a_{m,\text{next}} - a_{m,\text{now}})$, the source, with g and G the two declared coupling pairs at the body's cells (the capital G apart from the flux G_{ij} below, which always carries two Node indices; CARRIED [9, 8.5]). The scheme is the Euler–Lagrange scheme of a two-point Lagrangian, so it is symplectic and has the exact invariant

$$J = I_m + \alpha I_l + g \sum_{i \in R} D_i (a_{m,\text{next}} - a_{m,\text{now}})_i (a_{l,\text{next}} - a_{l,\text{now}})_i, \quad \alpha = g D_{\text{in}}/G, \quad (4)$$

I_m and I_l the two families' forms, D_{in} the weight on the body's cells and α one number of the body: conserved exactly on every board, before the remainders (PROVED [9, 8.5]). The amplitude form of the coupling is withdrawn as tachyonic, its mass matrix at zero wave vector having a negative eigenvalue.

The label matrix. Every record carries its rows per label value of each arm. A polar body with the integer axis $\mathbf{u} = (a, b)$ applies at its cells the integer matrix $\mathbf{M}_{\mathbf{u}} = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ to the record's pair of label rows, verb (B), with no division: the common factor $a^2 + b^2$ scales the record's form by Brahmagupta's identity $(ax + by)^2 + (-bx + ay)^2 = (a^2 + b^2)(x^2 + y^2)$, exact on integers [9, 9.16 (2)]. The polariser's angle is this integer axis; there is no table of angles [9, 9.14 (b)].

The flux. Let $e_i = D_i (a_{\text{next},i}^2 + a_{\text{now},i}^2) - \frac{1}{3} a_{\text{next},i} (\mathbf{A}\mathbf{a}_{\text{now}})_i$ be the Node's share of the form, so that $I = \sum_i e_i$.

Proposition 1 (the local flux identity). *Between two intervals,*

$$e_i(t) - e_i(t-1) = \sum_{j \text{ read by } i} G_{ij}, \quad G_{ij} = \frac{1}{3} A_{ij} (a_{\text{now},i} a_{\text{before},j} - a_{\text{before},i} a_{\text{now},j}), \quad G_{ij} = -G_{ji}. \quad (5)$$

Proof. Substitute the rule $D_i (a_{\text{next},i} + a_{\text{before},i}) = (\mathbf{A}\mathbf{a}_{\text{now}})_i/3$ into the difference of e_i between the two intervals [9, 9.19 (3)]. $\square$

G_{ij} is the flux into Node i from Node j : a bilinear form of the record's own two levels at the two ends of a Link, free of the pairs (they cancel), antisymmetric and local. For a character moving from j to i it is positive. The one-way inward flux into one Node, summed over intervals and Ports of $\max(G_{ij}, 0)$, equals the record's form over the passage of a packet (COMPUTED to six digits on a chain [9, 9.19 (3)]). The rectification $\max(G, 0)$ is the click's own comparison: a record's energy is counted where it enters.

The residue and the wheel. Two things are called a remainder. The rule's remainder r is kept per Node on the record's row and is the law's own. A record's residue u is one integer, read once: the rule's remainder at the record's birth cell at the interval of the click that births it; its wheel

is $W = 3 \text{ den} / \text{gcd}(\text{num}, 3)$ of that cell's pair for the clicking record's family, in lowest terms. No residue, seed, order or wheel is declared anywhere; the loader refuses them (ADOPTED, record 1872 [6]; [9, 9.19 (4)]). From $r = 0$ every remainder at a Node is a multiple of $\text{gcd}(\text{num}, 3)$, so at most $3 \text{ den} / \text{gcd}(\text{num}, 3)$ values occur (PROVED); that they are reached and equidistributed is COMPUTED, not proved. The granularity of Born's rule on a wheel of W values biases a share by at most $1/(2W)$, so every birth cell carries a pair with $W \geq 500$, checked at load [9, 9.19 (4a)]. With the law's remainder the residues are equidistributed and not a permutation, and the counts are binomial about the shares, as nature's are; the run stays deterministic and reproducible.

Definition 1 (The detector and the click). A *detector* is one connected named region, a cube of side at least 3 (a smaller cube, or one name on disconnected pieces, is refused at load); separate places are separate names [9, 9.25 (7)]; record 1899 [6]. A record carries a *ladder*, the detectors it names in a declared order with the face receiver last. At its birth the record's threshold is fixed, $\theta = (2u + 1)T/(2W)$, with T the record's form I , and its running total starts at $C = 0$. At every interval t each detector d of the ladder receives the increment

$$f_d(t) = \sum_{\substack{i \in d, j \notin d \\ j \text{ read by } i}} \max(G_{ij}(t), 0), \quad (6)$$

the one-way inward flux through the region's Ports from outside it (a Link between two Nodes of one detector is no Port of it), and $C(t) = C(t-1) + \sum_d f_d(t)$. The *click* falls at the first interval t^* with $C(t^*) \geq \theta$, that is $2W C(t^*) \geq (2u + 1)T$, at the detector whose segment of that interval's increment, laid out in the ladder's order, contains $\theta - C(t^* - 1)$; there the record ends, whole, everywhere, at that interval. The click is the detector's: it names the detector, never a Node inside it, and is reported per detector [9, 9.25 (2)]; record 1899 [6]. T and C are integers once scaled by 3 and by the pairs' common wall.

The three tests hold for the click: generic, the integers u , T , W and the fluxes, no family name; vector, sums and one comparison per detector, the division's comparison (D), no root; local, each detector's increment is its own Ports', and the record's two integers θ and $C(t-1)$ travel with the record as its residue does. What is not local is the deletion: the record ends at once, everywhere, the one non-local act of the law (the model owner's decision, record 1888 [6]).

Theorem 2 (one quantum, one click). *With the face receiver last on every ladder, every record clicks exactly once.*

Proof. C is non-decreasing and $\theta < T$, while everything that leaves the board offers its whole form at the face; so θ is crossed exactly once, and the deletion at that interval leaves nothing to cross it again [9, 9.25 (2)]. $\square$

Theorem 3 (Born's rule for a detector). *Let the residue u be spread evenly over $\{0, \dots, W-1\}$. Then the probability that a record clicks at detector d is $C_d(\infty)/T$, the detector's share of the record's total inward flux, whatever the flux's time profile, to the grain $1/W$; over the ladder with the face the shares sum to one. A detector's share is the sum of its Nodes' shares.*

Proof. θ is spread evenly over a grid of W points in $[0, T)$. The segments of the increments, one per interval and detector in the ladder's order, partition $[0, C(\infty))$ with the lengths $f_d(t)$, so the click falls at (t, d) with probability $f_d(t)/T$ to the grain $1/W$; summing over t gives $C_d(\infty)/T$. The total length of a detector's segments does not depend on the order within an interval [9, 9.25 (3), (4)]. $\square$

The residues of the law's remainder are equidistributed (COMPUTED), not a permutation, so the counts are binomial about the shares [9, 9.19 (4b)]. Two other forms were weighed and refused: a race in which each Node shouts on its own threshold is not Born's rule (two Nodes receiving the shares 0.9 and 0.1 give 17/18 to the first, PROVED), and exactness needs the chance normalised by the record's remaining norm, which is the increment ladder read cell by cell [9, 9.25 (5)].

Definition 2 (The birth). The reading $\mathbf{E}$ maps a record at a set of cells to its offer there; its transpose $\mathbf{E}^\top$ maps a number back to a record, the real part of a character written on cells [9, 9.13]. A birth is $\mathbf{E}^\top$ at a click, and only at a click. An *emitter* is a body of a massive family with a stock of M excitations: at interval 0 its first excited record is on the board, the body's mode at both levels; excited record k clicks at its own rung on its own centre cell, and at that click it ends and $\mathbf{E}^\top$ writes two records, the born record (one quantum) and, while the stock lasts, excited record $k + 1$ (one quantum); so quanta are conserved at every click, and the emission times are the rungs, the spontaneous emission of the algebra [9, 9.17 (2), (4)]. The born record's two levels are integers of the world written by the generator, on one cell the pair [now, -now], on a line of cells a travelling profile, copied once at the click and never driven afterwards [9, 9.17 (5), (6)]. A *crystal* is $\mathbf{E}$, the arriving record's click at its cells, then the split of the arriving clock into the two born clocks, then $\mathbf{E}^\top$, the birth of an entangled pair with its declared branches [9, 9.13]; record 1879 [6].

Where a tensor enters. In one place only: the pair, a record of fixed rank 2 whose label module is $\mathbb{Z}^2 \otimes \mathbb{Z}^2$, four integer weights on the joint labels, each arm's rows over the Nodes as any record's. It is required only where entanglement is measured, since a product state gives the CHSH sum at most 2 (PROVED: CHSH's bound holds for weights that factor). It is born only by the crystal during the run, never written at interval 0, and two separately born records are never joined; so the generator writes only vectors. The pair's click reads its joint weights $R(o_A, o_B) = J(o_A, o_B)^2$, $J(o_A, o_B)$ the joint amplitude of the outcomes o_A and o_B (apart from the invariant J of Eq. (4)), on the joint ladder of the four outcomes, the one residue choosing the outcome for both arms at once; the click of one arm alone reads the weights summed over the other arm's labels, the partial trace, kept nowhere [9, 9.22 (6), 9.25 (6)]; record 1880 [6].

Theorem 4 (no signalling through the click). *For a pair record, the share of records clicking + at Alice's detector is $|\mathbf{w}_+|^2 / (|\mathbf{w}_+|^2 + |\mathbf{w}_-|^2)$, with $\mathbf{w}_o$ her arm's label state projected on her own axis, exactly in integers and independent of Bob's axis, placement and distance; the same holds for Bob. So the counts at one detector do not depend on what is done at the other, and no information passes through the click, at any speed.*

Proof. Her marginal is $\sum_b R(+, b) / \sum R$, and $\sum_b R(o_A, b) = (a_B^2 + b_B^2) |\mathbf{w}_{o_A}|^2$ by Brahmagupta's identity, Bob's axis being (a_B, b_B) ; the scale $a_B^2 + b_B^2$ is common to both outcomes and cancels in the share. Her click's time is her own arm's flux and her outcome's boundary her own projection [9, 9.25 (6)]. $\square$

What is non-local is Bob's outcome given Alice's, decided by the same residue and the joint weights: the correlation, not a signal. A deletion that spread from Alice's click at one Link per interval would be a local rule at each arm, and every such rule gives the CHSH sum at most 2; so the deletion is at once, the algebra's one non-local act [9, 9.25 (6)]; record 1888 [6].

Theorem 5 (the direction of time). *Given the neighbours' reads, the step $(a_{\text{before}}, r) \mapsto (a_{\text{next}}, r')$ of one row is a bijection of $\mathbb{Z} \times \{0, \dots, 3 \text{ den} - 1\}$ onto itself, with the exact inverse $a_{\text{before}} = \lceil m / (3 \text{ den}) \rceil$, $r = 3 \text{ den } a_{\text{before}} - m$, where $m = \text{num } S_6 - 3 \text{ den } a_{\text{next}} - r'$. The coupled step is two such bijections in a fixed order, inverted in the reverse order. The click deletes a summand and has no inverse.*

Proof. $(a_{\text{before}}, r) \mapsto n = \text{num } S_6 - 3 \text{ den } a_{\text{before}} + r$ is a bijection onto $\mathbb{Z}$ by the uniqueness of the Euclidean division, and so is $n \mapsto (\lfloor n/(3 \text{ den}) \rfloor, n \bmod 3 \text{ den})$ [9, 8.8]; [9, 9.20 (A) 4]. $\square$

The Inside is reversible to the bit on every board and is not symmetric in form, the remainder living on the later level: the rule carries an arrow in its bookkeeping and none in its information. The arrow of time is born at the click and nowhere in the rule. The check that shows it is a property test and never a pin: a planted board with no detector run forward and back under the exact inverse returns bit for bit, and the same board with a detector does not, by the deleted record exactly [9, 9.20].

The families. A family is its vacuum pair $[\text{num}, \text{den}]$ and its label module ($\mathbb{Z}^2$ per arm for the hand, $\mathbb{Z}$ for a family with no hand), and nothing else [9, 9.22 (1)]. From the pair alone follow its rest frequency, $\cos \omega_0 = \text{num}/\text{den}$ (light has none), its operator block on an empty board and its band (Section 3), its stability, whether it can hold a body (only a massive family can), and its remainder's range. A born record's clock is a datum of the birth that makes it, not of the family. A world has at most three families, fixed by one pair per family per Node: light, the experiment's matter and the holder of the tools' bodies (ADOPTED, record 1875 [6]).

The experiment as data. An experiment is a list of what is on the board, and nothing else [9, 9.22 (2)]; records 1875, 1882 and 1885 [6]:

- (i) the board: the extents per axis, and per axis periodic or open, an open axis carrying the face receiver at its border, one border for every family;
- (ii) the families, at most three, each its pair and label module;
- (iii) the material map: which Nodes carry which pair per family, which a coupling (g, G) , which an axis (a, b) , every region a cube by its vertex and edge;
- (iv) the occupation: which bound bodies hold quanta and how many, at which amplitude; which packets are on the board; which bodies carry a stock and birth which family with which born profile; which body births a pair, with the crystal's branches;
- (v) the momentum: every entry carries its total momentum, the wave vector $\mathbf{K}$ of its character, $\mathbf{0}$ at rest; there is no acceleration, a moving entry is written moving at interval 0 and its energy of motion is part of the initial state (record 1884 [6]);
- (vi) the names: the detectors' regions and each birth's ladder, the face receiver last.

Not part of it, and refused at load: a residue, a wheel, a seed written by hand, a rate, a train, a heading, a take, a grace, a table, a per-family border, a ramp.

The initial state. The initial element is

$$\mathbf{v}_0 = \sum_{\text{occupied bound modes}} (\text{the quanta}) \times (\text{the mode's integer profile at the declared amplitude, at both levels}) + \sum_{\text{packets}} \quad (7)$$

with every remainder 0 and nothing else on the board: no light record and no remainder at interval 0 [9, 9.22 (3)]. The modes are those of the composed operator of the body's family, every region in place. The generator is the board's own operator iterated: the power iteration $\mathbf{v} \mapsto (\mathbf{M} + 2\mathbf{1})\mathbf{v}$, $\mathbf{1}$ the identity, applies the law's own read and nothing else and converges to the bound mode from any start, $\mathbf{M} + 2\mathbf{1}$ having nonnegative entries (PROVED by Perron–Frobenius); in integers it is reproducible bit for bit [9, 9.22 (7)]. So the law *steps* the operator, the generator *relaxes* with it, and the click compares and deletes; nothing on the board relaxes.

The load check. The initial element is written once into the experiment's input file, with each occupied mode's clock as a rational a/b , and the engine checks it at load in integers only and says

LAWFUL or REFUSED with the reason: the law and generator the file was made under (a change of the laws forces regeneration); the eigen-equation's residual at every Node of the profile $\mathbf{p}$,

$$|b \text{ num}_i (S_6 \mathbf{p})_i - 3 \text{ den}_i a p_i| \leq b (3 \text{ num}_i + 6 \text{ den}_i), \quad (8)$$

PROVED as the bound for the rounded profile of an exact mode; a/b above the band's top and below 2; every remainder 0; the stability of the composed operator; the rich birth cells; the detectors' cubes; and the refused keys [9, 9.22 (3), (7)]; record 1886 [6]. Recomputation at load is a diagnostic only, since a host's float mode computation is not reproducible across hosts.

3 The massive family: the band, the cone, the bound body and the moving packet

The rule of Eq. (2) gives every family one declared pair and lets the one operator carry mass, a clock and a body [9, chapter 8, 9.24]. This section states what follows from the pair before any run; each sentence carries the algebra document's mark, and every number is a COMPUTATION declared before its run.

The band and the pitch. Read on a character $\cos(\omega t - \mathbf{k} \cdot \mathbf{x})$ of a periodic board, with ω the frequency angle per interval and $\mathbf{k}$ the wave vector (angles k_x, k_y, k_z per Link), Eq. (2) without its remainder holds exactly when $2 \cos \omega = (2 \text{ num}/(3 \text{ den}))(\cos k_x + \cos k_y + \cos k_z)$; with light's band $\cos \omega_l(\mathbf{k}) = (\cos k_x + \cos k_y + \cos k_z)/3$ at the same $\mathbf{k}$ the same line reads

$$\boxed{\cos \omega = \cos \omega_0 \cos \omega_l(\mathbf{k}), \quad \cos \omega_0 = \text{num}/\text{den}}, \quad (9)$$

the massive band as light's band times the cosine of the pitch ω_0 , the family's rest frequency (PROVED for every periodic board [9, 8.1]). At $\mathbf{k} = \mathbf{0}$ the gap is the pitch, the rest period $N_0 = 2\pi/\omega_0$ intervals and the rest energy $h \omega_0$, h the action; light has $\omega_0 = 0$, a zero mode that is a level and not a clock, so the whole difference between light and a massive family in the algebra is a gap. With $2 - 2 \cos \omega = 4 \sin^2(\omega/2)$ the band is light's form with the pace squared scaled by num/den , one substitution.

The two paces and the exact cone. Near $\mathbf{k} = \mathbf{0}$ light's band gives $\omega_l^2 = k^2/3 + O(k^4)$, k the length of $\mathbf{k}$: the pace of light $c = 1/\sqrt{3}$ Links per interval. To second order in ω and $\mathbf{k}$ together the massive band is Klein–Gordon's with the cone $c_m^2 = \cos \omega_0 c^2$, a massive record's limiting pace c_m below light's by $\omega_0^2/4$ (0.56 percent at the pair [800, 809], $N_0 = 42.08$, COMPUTATION); with no truncation in ω_0 , at the band's bottom,

$$\boxed{\omega^2 = \omega_0^2 + c_{\text{eff}}^2 k^2 + O(k^4), \quad c_{\text{eff}}^2 = \cos \omega_0 \frac{\omega_0}{\sin \omega_0} c^2, \quad \frac{c_{\text{eff}}^2}{c_m^2} = \frac{\omega_0}{\sin \omega_0} = 1 + \frac{\omega_0^2}{6} + O(\omega_0^4)}, \quad (10)$$

c_{eff} the effective pace of the band's bottom (PROVED as the expansion there [9, 8.1]; 1.0037 for the ratio at $\omega_0 = 0.15$, COMPUTATION).

Light against itself. Light's own band, the pair [1, 1], disperses by wavelength and by direction with nothing declared: along a unit direction $\mathbf{n}$ with $|\mathbf{k}| = k$, the group velocity v_g obeys

$$\boxed{\frac{v_g}{c} = 1 - \frac{(3 \sum_i n_i^4 - 1) k^2}{24} + O(k^4)}, \quad (11)$$

the coefficient 0 on a body diagonal, 1/2 on a face diagonal, 2 on an axis and 4/5 averaged over the sphere (COMPUTATION, the series checked against the exact group pace [27]): a prediction of form, the quadratic dispersion of light anisotropic with the cubic pattern of the 48 and no free coefficient beyond the one scale; nature today has bounds and no measured coefficient.

The bound body at rest. A body is no new object: a region R of the operator, a cube of Nodes on which the 48 act, where the massive family's pair is $[\text{num}', \text{den}']$ with $\text{num}'/\text{den}' > \text{num}/\text{den}$, a well [9, 8.3]. Its clock is the composed operator's bound mode, the angle ω_b below the rest frequency ω_0 , whose $2 \cos \omega_b$ lies above the vacuum band's top; the mode's tail falls exponentially away from the well (PROVED), so the torus must be large enough that its wrapped tail stays below half a unit at the amplitude [9, 9.22 (9)]. A bound tool body is of the holder family, distinct from the experiment's matter (record 1875 [6]). At interval 0 the body's mode is written by the generator at both levels, and the load check verifies it (Eq. (8)).

The moving body. The one operator is linear in the state and, away from the regions of material, commutes with every translation.

Theorem 6 (no self-bound body). *A linear translation-invariant map has the characters as its modes and nothing else stationary, so a localised stationary state needs a region of material that breaks the translations. There is no self-bound localised body in the one algebra: a moving body is either a packet of a family carried by the character of its momentum, or a region of material, which does not move.*

The proof is [9, 9.24 (1), (3)]; a well carried by a record of the holder family would need a term of degree three in the amplitudes, refused by the three tests. On a homogeneous board every character is a mode with $3 \cos \omega = (\text{num}/\text{den}) \sum_i \cos k_i$, and a packet, a sum of characters, is an exact solution, each character advanced by its own clock; the written initial state, the profile times the character of $\mathbf{K}$ at both levels, moves from interval 0 with no ramp (PROVED [9, 9.24 (2)]). Its group velocity per axis is $v_i = (\text{num}/\text{den}) \sin k_i / (3 \sin \omega)$, and at small $\mathbf{K}$ and ω , $\omega^2 = \omega_0^2 + k^2/3$ with $\omega_0^2 = 2(1 - \text{num}/\text{den})$, the relativistic band with $c = 1/\sqrt{3}$ and the mass ω_0 . At the packet's centre, which moves at $v = v_g(\mathbf{K})$, the phase advances by $\omega(\mathbf{K}) - \mathbf{K} \cdot \mathbf{v}$ per interval, so

$$\boxed{\frac{f}{f_0} = \frac{\omega(\mathbf{K}) - \mathbf{K} \cdot \mathbf{v}}{\omega_0}}, \quad (12)$$

f_0 the rest tick and f the moving packet's own tick: time dilation from the dispersion, with no rule for it, exactly the Lorentz factor's inverse for the relativistic band and, on the lattice band along an axis at $v = 1/3$, the tick ratio 0.8146 (the pin of [9, 9.22 (8)], rows 5, 11 and 12; [10, B]). A packet of width σ (Links) spreads on the time $2\sigma^2/\omega''$, ω'' the curvature of the dispersion at rest, so a moving clock is a large packet, about a hundred Links wide for the experiments' holds [9, 9.24 (2)]. A detector or an emitter carried by a moving body is a *moving name*: its cells translate at the body's own group velocity from its $\mathbf{K}$, with no declared rate or heading; the click reads the flux into the moving region and the birth writes on it (the model owner's decision, record 1889 [6]; [9, 9.24 (4)]). There is no acceleration during a run: a moving entry is written moving at interval 0 and its energy of motion is conserved with the rest (record 1884 [6]). On the board's own surface nothing contracts: the profile is as written. The model has clocks and no rods: a moving rigid arm is not yet defined, and a light clock moving across its arm reads the packet's own dilation; the longitudinal clock's factor is not claimed (record 1895 [6]; [9, 9.24 (6)]).

The coupling and the index. A body coupled to light by the pairs g and G (Eq. (4)) scatters light at its cells; a medium body at rest has the index n with $n^2 = 1 + Gg/(\omega_0^2 - \omega^2)$ in closed form,

the design's, its ω_0 the medium's pitch and ω light's frequency; the medium is a bound body of the holder family, its pin re-derived blind on that body (record 1877 [6]; [9, 9.22 (8)], rows 14 and 15).

The push. A body's cells stepping by its momentum, with its stress the flux of momentum at its outer Ports, is CARRIED from its design and not proved in the one algebra; momentum at a body is conserved by declaration, not by the algebra, and a push from the law's own values is OPEN [9, 8.11, 9.21 (6), (7a)]. Every result this paper states as proved holds without it.

Known formulas and predictions. Every experiment of Section 8 carries (K), a known-formula row, whose algebraic identity is a textbook formula compared with, or (P), a prediction, a number the algebra gives and no known formula gives; a prediction is not a pass. The predictions of the massive family, each with its falsifier: the two-pace world, a massive record's limiting pace c_m below c (Eq. (10)); light's quadratic dispersion with the cubic pattern (Eq. (11)); and a moving clock's tick from the lattice band, Eq. (12), whose departure from the continuum's $1/\gamma$ (γ the Lorentz factor) is the lattice's own signature, read in the bound clock's row [10, B].

4 From click to click

Everything compared with nature is a click, a count of clicks or a ratio of intervals between clicks; nothing inside the board is compared. This section is the chain from a birth to the counts, and the two theorems of the Outside.

The steps between two clicks. Every comparison of this paper walks one chain of six steps.

- (1) The birth: $\mathbf{E}^\top$ at an emitter's click, the born record's two levels written once on the emitter's cells; its residue u and wheel W read from the law at the birth cell (Definition 2; DECLARATION for the born profile, the law for the rest).
- (2) The law: the one operator, at most one Link per interval, the family's pair at every Node, the coupling and the label matrix at the tools' cells (Eq. (1)).
- (3) The click: the record's one-way inward flux into the detectors named on its ladder crosses its threshold, and the record ends whole, everywhere, at once (Definition 1; DETECTOR).
- (4) The output: each click's detector and interval, as the experiment's output file gives them (record 1887 [6]; DETECTOR).
- (5) A count of clicks per detector, or a ratio of intervals between clicks (COMPUTATION on clicks; nothing of the board enters).
- (6) The identity the algebra states for that count or ratio, the form of nature compared with.

Step (6) is the only place where a formula meets a measurement. Anything read from the board itself is a GameBoard diagnostic and enters no comparison.

Locality and discreteness Outside, two theorems. Two properties of the game above the board follow and are not assumed.

Theorem 7 (Locality Outside). *Every step of the law but the click reads a Node's own record and its six neighbours, so a change at one Node reaches Manhattan distance m at interval m and no further [9, 9.20 (A) 5]. The click is the one non-local act: the record ends at once, everywhere. The information a click yields Outside is its detector and its interval; which detector clicks is decided by the record's own residue and its own flux, and by Theorem 4 no count at one detector depends on what is done at another. So no information passes Outside faster than one Link per interval.*

Theorem 8 (Discreteness Outside). *The counts of one detector form a free abelian monoid and their ratios lie in $\mathbb{Q}$: every Outside quantity is an integer count or a ratio of counts. A click names a*

detector, a cube of side at least 3, never a Node inside it; two places read are two detectors, separate names. A velocity read across Δx Links and Δn intervals is the rational $\Delta x/\Delta n$; and c Outside is a bound, at most one Link per interval on every line, with $1/\sqrt{3}$ the band's pace as the wave number goes to 0 (Eq. (9)). Nothing continuous enters above the board.

Neither is an added assumption: the first is the law's locality with the click's decision (record 1888 [6]), the second the click's definition with the detector's cube (record 1899 [6]). What is more than these is not a theorem: that the apparatus is itself inside the law, emitters and detectors regions of the one element, is the model's construction, and the paper claims only what the clicks read.

Lorentz Outside. A boost, which mixes a step in space with a step in time, is not among the 48, and no linear bijection that preserves the polyhedral cone $|\mathbf{x}|_1 \leq t$ is a boost. Lorentz's form is the symmetry of the clicks Outside, assumed of the world and arrived at from the two assumptions of the click theorem [50]; the law itself derives a moving clock's slowing from its own dispersion, Eq. (12), and nothing more of the Lorentz group.

5 From the algebra, the physics

What follows from the algebra is stated first in one paragraph, each item with its word, and then subsection by subsection.

What is proved. Exact on the GameBoard, no experiment needed [9, 9.19, 9.20 (A), 9.25]: the group of the six Ports and, within it, the group of order 24 (Theorem 1); the equivariance of the law under the 48 and the translations, a world breaking the symmetry only through its declared data; the content, the number of quanta, conserved between clicks; per family the form I and for two coupled families J , conserved exactly before the remainders, with the exact remainder identity; the local flux identity (Proposition 1); the step's bijection, the click the one deletion (Theorem 5); locality, one Link per interval; the stability condition, every mode a rotation exactly when the largest eigenvalue is below 2; one quantum, one click (Theorem 2); Born's rule for a detector to the grain $1/W$ (Theorem 3); no signalling through the click (Theorem 4); a product state gives the CHSH sum at most 2; the CHSH sum $14/5$ on the integer axes of the Bell experiment (Section 5.4); Malus's share $a^2/(a^2 + b^2)$ on an integer axis (a, b) ; the band, the cone and light's dispersion (Section 3); no self-bound body, and a moving packet's tick from its dispersion (Theorem 6, Eq. (12)). Recovered as the wave number goes to 0: $c = 1/\sqrt{3}$, isotropic at second order by the 48; Klein–Gordon's dispersion; the continuum's $1/\gamma$. Computed and not proved: the equidistribution of the law's remainders. Carried and not proved: the push. Reached under a named assumption: Lorentz's form of the clicks Outside [50].

5.1 Geometry and symmetries: the directions and $c = 1/\sqrt{3}$

The symmetries. The law is covariant under the translations of the torus and of the interval and under the 48 signed permutations of the axes: for every g in the 48 and every translation, $\mathbf{S}(g \cdot w, g \cdot \mathbf{v}) = g \cdot \mathbf{S}(w, \mathbf{v})$, w the experiment's declared data; the click commutes with g in the same sense, the detectors' cells moving with the world, provided the order within an interval is data of the world [9, 9.19 (6), 9.25 (4)]. A world breaks the symmetry only through its regions of material. The 48 are the GameBoard's stand-in for the rotation group; a boost is not among them. The six neighbours of a Node are the vertices of the octahedron $|x| + |y| + |z| \leq 1$, the causal front of one

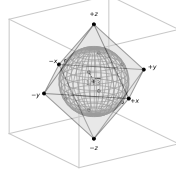


Fig. 2 The Nodes one interval from a Node: the six neighbours are the vertices of the octahedron $|x| + |y| + |z| \leq 1$, the causal front of one interval; its inscribed sphere, of radius $1/\sqrt{3}$, touches the eight faces on the cube diagonals, and $1/\sqrt{3}$ is light's band pace as the wave number goes to 0 (Eq. (9)); the faint cube shares the 48 signed permutations of the axes. Drawn by `octahedron.py` from the definitions; no run.

interval; its inscribed sphere, of radius $1/\sqrt{3}$, touches the eight faces on the cube's diagonals, and $1/\sqrt{3}$ is light's band pace as the wave number goes to 0 (Eq. (9)).

5.2 Conservation, the flux and momentum

The books. Between clicks nothing but a birth changes the summands, so the content is exact; a click deletes one quantum and a birth writes one, and at an emitter's click the photon and the next excited record are written as the excited record ends, so quanta are conserved at every click [9, 9.17 (4), 9.20 (A) 3]. On an open axis what leaves clicks at the face receiver and is counted there; every record clicks.

The continuity equation, exact. Summed over a region, the local flux identity (Eq. (5)) loses every Link inside the region by the antisymmetry $G_{ij} = -G_{ji}$: the change of the region's share of the form in one interval is the flux through its Ports. This is the lattice form of the continuity equation, exact in the real reading, with the remainders' exact identity beside it (DERIVED from Proposition 1).

Momentum. On a homogeneous board every character is a mode of the operator, so its wave vector is exact (PROVED [9, 1.6]); at a body the difference goes into the body by the push, CARRIED (Section 3).

5.3 Measurement and the quadratic form

The reading is quadratic by construction. The click reads the one-way inward flux, a bilinear form of the record's own two levels at the two ends of a Link, against the record's conserved form I , a quadratic form (Eqs. (3) and (5)); Born's rule, a detector's share of that flux, is a theorem (Theorem 3) and not an axiom. No principle below the square is offered: the square is the form the law conserves. A static level has $I = 0$ and clicks nowhere.

Interference, by the formulas alone. The law is linear, so two paths of one record add or cancel at a Node; a detector counts the record's inward flux, so where the two paths cancel no flux enters and no click comes. At the exit ports of a balanced Mach–Zehnder interferometer with two large detectors, one region each, the dark port has no inward flux and never reaches a rung, and the bright port takes every click, against a half-and-half split without interference (record 1891 [6]). On a screen of many detectors the counts are binomial about each detector's share of the record's total inward flux (Theorem 3); the two slits' visibility is re-derived blind on the grown board (Section 8).

Malus’s share. A record with the label state H arriving at a polariser with the integer axis (a, b) clicks at the detector along the axis with the share $a^2/(a^2 + b^2)$, by the label matrix and Brahmagupta’s identity [9, 9.16 (4)]: at the axes $(1, 1)$, $(5, 1)$, $(15, 8)$ and $(3, 2)$ the shares $1/2$, $25/26$, $225/289$ and $9/13$, at 45 , 11.31 , 28.07 and 33.69 degrees. There is no table of angles.

de Broglie and the rest energy. A matter record’s wave number is the band’s $\mathbf{k}$ at its frequency (Eq. (9)), the de Broglie fringes’ spacing its consequence; the rest energy is $h\omega_0$, the pitch (Section 3).

5.4 Bell and CHSH: from weights to counts

The source. The pair is born by the crystal: the arriving record’s click at the crystal’s cells, then $\mathbf{E}^\top$, the birth of one record of rank 2 in the state $HV + VH$, its two arms going to two polarisers with their detectors (Definition 2; record 1879 [6]). One residue u per pair chooses the outcome for both arms at once on the joint ladder of the four outcomes, the one non-local act.

The weights, exactly. With Alice’s axes $(1, 0)$ and $(1, 1)$ and Bob’s $(1, 2)$ and $(3, 1)$, the joint weights of the outcomes $(++, +-, -+, --)$ are [9, 9.16 (4)]:

Alice’s axis, Bob’s axis	the weights	the correlation E
$(1, 0), (1, 2)$	4, 1, 1, 4 of 10	$3/5$
$(1, 0), (3, 1)$	1, 9, 9, 1 of 20	$-4/5$
$(1, 1), (1, 2)$	9, 1, 1, 9 of 20	$4/5$
$(1, 1), (3, 1)$	16, 4, 4, 16 of 40	$3/5$

so the CHSH sum is $S = 3/5 + 4/5 + 4/5 + 3/5 = 14/5$ exactly. A product state gives $S \leq 2$ (PROVED, CHSH’s bound for weights that factor), and $2\sqrt{2}$ (Tsirelson’s value, nature’s optimum) is not a pin [9, 9.22 (6)].

The counts as distributions. With the residue the law’s own remainder, the counts are binomial about the weights [9, 9.19 (4b)]. At $n = 100$ pair records per setting the expected counts are $40/10/10/40$, $5/45/45/5$, $45/5/5/45$ and $40/10/10/40$, and $S = 2.80$ with the standard deviation 0.141 , the variance $\sum(1 - E^2)/n = 2.00/n$ over the four settings. The pin, written before any run: S in $[2.38, 3.22]$ (three standard deviations), the falsifier $S \leq 2$.

The marginals and the order of the outcomes. Each party’s share of $+$ is $1/2$ at every axis for the state $HV + VH$ (COMPUTED), and by Theorem 4 it does not depend on the far axis, exactly in integers; each party’s counts are binomial about it. Alice’s outcome is chosen on the outer segments of the joint ladder, whose union does not depend on Bob’s axis, so her record-by-record outcomes and click intervals do not depend on his setting (the property test’s seventh test [9, 9.25 (8)]). Whether the order of Bob’s outcomes, chosen on the inner segments, carries Alice’s setting under the law’s remainder is to be re-derived; this paper claims nothing for it.

6 The GameBoard: the way the algebra is computed with

The GameBoard is the way the algebra is computed with, not the object: a system of information transfer with costs, read in two ways, the detector’s click a measurement and the host’s view a diagnostic. This section states the law as one map with its postulates, the state it acts on under the record kinds of Sections 2 and 3, the six operations as the engine runs them, the read-out, the code and the ledger; every sentence is the algebra document’s [9, chapters 2, 3 and 8]. The law as the

engine ran it before 2026-09-23, the rows that hop whole with their wheel, their rungs and their components, is its record [52].

6.1 The law and its implementation

The one map. The law is the map of Eq. (??) applied at every Node at every interval, per component s of the state vector, each with its declared rate r and wall d : the component is translated by its rate, counted in walls, capped at $a = 1$ where a cap is declared and uncapped elsewhere (the whole part $\lfloor |s|/d \rfloor$ with the sign), and reduced by the walls it holds; every wall counted is an event, and nothing else happens. One declared variant of the count, the centred step, takes the nearest whole number the accumulator holds, $(|s| + \lfloor d/2 \rfloor)$ divided by d with the sign; it is named where a rule uses it. Where the rate is a constant of the world (the linear block) every component has the closed form $s(t) = \lfloor s_0 + rt \rfloor$, r the rate as a fraction of the wall, so a record's state at any interval is a formula of its birth and its age; where the rate is a function of what arrives (the feedback block: the push, the drive, the coupling of the two kinds) the map is iterated and its continuum limit is the differential equation $ds/dt = \mathbf{r}(s)$. The read-out is one comparison, the click [9, chapter 2].

Rules, parameters, initial conditions. The law is stated as rules, apart from the parameters and the initial conditions a world supplies. The rules: (P1) every quantity of the state is a bounded integer and every operation integer arithmetic, no real number, root or floating point at run time (the boundedness a property of the declared worlds over their declared ticks, guarded by the run-time amplitude bound: the light pair puts the pace at the six-neighbour rule's causal limit, $c^2 = 1/3$, so on a full three-dimensional board the two modes $\mathbf{k} = 0$, a uniform level, and (π, π, π) , the checkerboard, are marginal, double roots, growing at most linearly if inserted, and no GO world inserts either, and what a sharp edge excites of them on a full board the run-time bound watches; on a chain or a layer the band edge lies inside the band); (P2) space is a cubic array of Nodes, each joined to six neighbours by Links through directional Ports, a Node's complete local information its NodeState, time a sequence of intervals; (P3) the rates and walls of the map are made of the six operations of Section 1 and nothing else; (P4) a rule reads its own record and the six neighbours; nothing is kept at a Node beyond the events there, no register or draw, and the remainder r of Eq. (??) is a component of the record's own state (the pair travels with the record [9, 8.8]), not a register at the Node; (P5) no row or body crosses more than one Link per interval; (P6) a record is read out once, by one comparison, at the click, the one step that deletes, and the click of a pair is one gather of one record from both settings, the law's one non-local operation. Beside P6, once: the pair is one record with two arms (Definition 1, the record $\psi = (00) + (11)$ of rank 2, each arm carrying the weights, nothing of one arm written on the other); the completion is one gather at the completion interval, no interval crosses between the arms and no arm's history is altered; the counts' no-signalling follows from the table's symmetry (Theorem ??); the completion is an operation of the law that relaxes P4 at that one step only, said so and not hidden. The parameters (P7, P8): the circle N_ϕ and its tables at $1/256$, the pair [num, den] of each record kind with its pitch, light the value [1, 1] (Eq. (??)), the rung W of each detector, the cells of each block with its lowered pair, the take of each absorbing object, and the roundings declared at load; the initial conditions: the box and its boundaries, the blocks with their momenta, the lamps and the detectors of the world file, and the seed, the bound mode's integer profile over the whole board, the engine reading integers [9, 8.7]. The law of this paper is the runtime at the commit Appendix B names. The rules chosen among few (P9) are assumptions, not derivations, each named where it enters and each marked in the algebra document as carried, proved or computed and not proved: the rule with the pair per kind, carried from the design, its band and its two paces proved; the click's definition at the rung,

carried on the author's word; the one formula of a block's motion, carried from the continuum, its two limits proved; the stress of light at a block's outer Ports as the wave's momentum flux, the design's declaration; and the bijection of the interval, the design's proof carried (Section 3). One axiom of the apparatus (P10): the click stores one pointer per set, the least-rank member of the family Theorem ?? forces, which is Born's form with the fundamental harmonic; which harmonic is a relabelling of the declared rate, and the harmonic constants are not derived. One assumption of the frame (P11): the GameBoard, Inside, is read only through its emitters and detectors, Outside; every result Outside is a click, a number of the board's state is never a reading, and every comparison with experiment is a comparison of clicks with readings; a measurement is itself an event of the board, an emitter putting a record on it and a detector ending the record at the click, the one step that deletes Inside and the one row that Outside gains. Nothing outside this list is assumed.

The state. The GameBoard is a box of Nodes in $\mathbb{Z}^3$, each joined to six neighbours by Links through six Ports, an axis periodic or open; time is the interval $t \in \mathbb{Z}$; the state at an interval is a vector $\mathbf{s}$ of bounded integers on a product of circles and lines. A Node's complete local information is its NodeState: per record kind present, the record's row, its two levels and its remainder $(a_{\text{now}}, a_{\text{before}}, r)$ of Eq. (??); at a block's cell, the block's lowered pair, its pointer in the units of the conserved form and its count; nothing else, no register, no draw, no field beyond the records there; beside the click's gather the model declares two other non-local objects, a body's cells acting as one (the foreign object's extent, an input of kind 3, a detector of itself across its own cells) and the record's own ledger (its pointer per cell, its u , its norm), the record's state travelling with it and not a Node's, so that a Node holds nothing but the rows present at it and the locality claim stops at the Inside's verbs [9, 2.11, 8.1, 8.6]. A block carries its cells, its momentum $\mathbf{p}$ in $\mathbb{Z}^3$ and its step's accumulator per axis with its remainder (Section 3). This paper states one law, the record kind; the rows that hop are its record. The rows that hop, a tuple (Node, direction, age, phase, emitter, content, record, label, amount, multiplicity) moving one Link per interval on a digital line, are the engine as built before 2026-09-23, kept as their record [9, 2.11]: Definition ?? and the theorems stated on them (Theorems ??, 5, ??, ?? and ??) stand as proved, on the objects they name.

The six operations as the engine runs them. The order of the six verbs within one interval, one line each in symbols [9, 2.1 to 2.6, 2.11], and what is not one of them (a root of the state at run time, a float, a true division, a draw) are in the records file [52].

The three tests, rule by rule. A rule enters the law only if it is generic (one primitive with declared integers, no family name or kind, the engine branching on no name), vector (one of the six verbs on the state vector, its rate at most bilinear in the state, no root, no float, no rounding at run time beyond the ones declared at load) and local (its own record and the six neighbouring Nodes, fixed work and storage for a fixed K , nothing kept at a Node); generic means commuting with the 48 [9, 2.8]. Each rule of Section 3 passes the three and carries its mark [9, 8.10]: the rule with the pair on the six reads, (B), (T) and (D), the band and the two paces proved, the range-one bound computed and not proved; the conserved form, a bilinear form of the two levels, proved on every board with the remainders' identity; the well of the pair on declared cells, the threshold and the tail computed and not proved, the norm bound proved for the declared pairs whose band lies strictly inside the stability limit ([800, 809] bounded); under [1, 1] the $k = 0$ and (π, π, π) modes are marginal and grow linearly, neither inserted by a declared world, P1's boundedness per declared world; the motion, the one formula carried, its two limits proved; the coupling, (B) linear in the other kind's two columns and one (D) per row, the Euler–Lagrange scheme and the invariant J with its remainder identity proved, the index and the pump computed and not proved; the click

across the block's cells at W , (E), the bilinear form and (D)'s comparison, the definition carried; the take, a declaration, no identity; the seed, the standing start proved; the direction of time, (D) with the ceiling, the bijection carried and its inverse exact; the block's momentum, step and push, the cadence and the bound proved as identities of the accumulator, the stress carried as the design's declaration; the force between two blocks through light, the closed form carried, on the GameBoard computed and not proved.

The read-out. The click is the evaluation (E) of a body's own record on its cells at the declared rung W : light arriving at the block's cells drives the block's record through the coupling, the block's pointer accumulates its own record's motion across all its cells in the units of the conserved form, and the click is the comparison $W \times (\text{the pointer}) \geq (\text{the record's norm})$, integers both, in the block's own clock (its own record's cycles across its cells, not a count of clicks); the click line carries the block's own count and the light record's birth stamp. Light clicks only at a detector set, a declared set of Nodes with its rung W ; at a massive body's cells light does not click but drives the body's record through the coupling, and the body's own click reads that; a free Node has no rung, no count and no reading; the verbs are lossless, light passes on, and the click needs no sink; where a world must absorb it declares a take, a damping pair on light's row at the cells of an object declared absorbing, a declaration of the world and not the law's [9, 8.6]. A click is an action, not a passive read: the record ends at the detector, its content enters the detector's own record, and the detector's count advances, the click stamped with that count [9, 3.1]. Between a birth and a click the interval is a bijection, its inverse exact (Section 3, the direction of time; for the rows that hop, Theorem 5); the click is the one deletion, and the pair's gather the one non-local step (P6).

The code as the law's runtime. Every registered world is run from its world file with its pin written before the run and its source fingerprint (Appendix B); the books balance at every interval in every run; the design's scripts that gate the code, each printing its record beside the design and none an engine run, are listed in the records file [52].

The ledger. The forcing ledger, in the paper's records file [52], is the paper's spine: for every result, the rules of Eq. (??) it starts from, the assumptions added beside them with their kind, the freedom left, and the ground of the derivation [34, 2026-09-22]: exact on the GameBoard (no limit, no average), a limit of the grain taken on paper with a rate, or a limit under a spatial condition, an average that covers the shell (over the shell's Nodes, over the directions of a fan declared in a ball, or over the sweep of an orbit), the condition then part of the claim. A result whose only reading is of the GameBoard's state is not in the table, and a result resting on a hypothesis beside the law or conjectured from the algebra stands in it only with that ground named in its row (the bending's coefficient, conjectured); the hypotheses beside the law are on the tree (Section 4).

7 How the algebra was reached: from the GameBoard to the group

The order of this paper is not the order of the finding. The object of Section 2 was reached from the GameBoard: from one Node and its six neighbours, one message per Link per interval, through the simulator's runs, in dated steps; what did not converge to the group and the ring was not put in. This section states the one choice and what it forces (the algebra document's chapter 7 [9]), the road from the operations to the group, the dated steps, the platform on which a formula is arrived at by two roads, and the check of the roads; the two direction tables, the simulator's part and the dictionary of the rows that hop, as the paper carried them, are its record [52].

The one choice, and the seven Nodes. The one choice of the model is the six Ports of a Node, $P = \{\pm x, \pm y, \pm z\}$ with the opposite involution, the L^1 neighbourhood of the cubic lattice: a Node and its six neighbours, the seven Nodes of the causal front, one message per Link per interval in each direction (Section 2). Two more choices, and no fourth: the phase circle $\mathbb{Z}_{N_\phi}$, N_ϕ per world, and the translation group of the torus, $\mathbb{Z}_X \times \mathbb{Z}_Y \times \mathbb{Z}_Z$, each factor a circle or a segment; the point group is then forced and is already the largest, the translation group is the board, and the only freedom that changes the physics is the neighbourhood, through the front, c and the tables [9, chapter 7].

What the choice forces: the 48, and the 24 within them. What is not chosen is the group of 48: every map that preserves the six Ports as a set and the lattice's Links is one of the signed permutations of the three axes, $3! \times 2^3 = 48$, and every one of them does; no three-dimensional lattice has a point group of order above 48 [9, chapter 7], and the cubic lattice's is the full octahedral group of that order, so this lattice keeps the 48 and no more. Nothing was added to it: the mass, the charge and the other contents of the family table live on the amounts, on which the 48 act trivially. The only $\mathbb{Z}$ -linear bijections that preserve the cone of one Link per interval are these 48; no boost preserves the polyhedral cone $|\mathbf{x}|_1 \leq t$ (whose linear automorphisms are finite), so no boost is among them, and Lorentz's form is reached Outside from the clicks (Section 4) and not as a symmetry of the lattice. The determinant splits the 48: the 24 rotations, the group of order 24, isomorphic to S_4 on the cube's four body diagonals, and the 24 reflections; the hand, needed to carry spin and polarisation, is the pseudoscalar that tells them apart, $h \mapsto \det(g) h$ (Theorem 1); the 24 are the octahedron's rotations, not a numeral chosen. The GameBoard showed that everything converges to this group and this ring, and what did not converge was not put in (the author's sentence, record 1109 [6]).

From the operations to the group. The operations did not deduce a group; they came first, as acts on the board: a row flies one Link per interval along its direction, rows meet at a Node, a body steps by an accumulator against a wall, a phase turns, identical rows merge, a detector clicks; each was then written as one of the six operations on a state vector. The group is the answer to which maps preserve what the operations act on, a Node of six Ports and its Links and a cone of one Link per interval: the bijections of the Ports that keep opposite Ports opposite are the 48 signed permutations, and they are the only $\mathbb{Z}$ -linear bijections that preserve the cone, no boost preserving the polyhedral cone $|\mathbf{x}|_1 \leq t$ (whose linear automorphisms are finite), so Lorentz's form is reached Outside from the clicks and not as a symmetry of the lattice (Section 4). The 48 first appeared in the code as the invariance of one operation, the collision's table, and were recognised after as the symmetry of all of them; the 24 came from the hand, the pseudoscalar that carrying spin and polarisation needed, which $\det(g)$ keeps or flips. The other objects came the same way: the step and the flight are shifts, hence the translation group; the turn adds modulo N_ϕ , hence $\mathbb{Z}_{N_\phi}$, and the merge of records is the addition of $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$; the collision is a permutation of a finite set; and c was not chosen, it is the flight operator's norm. What the group does back is to force forms: a rule that reads only the local state is required to commute with the 48 (the P-rule, a requirement beyond locality), and the symmetric integer matrices fixed by the 48 are the multiples of the identity, so the only isotropic quadratic reading of a body's momentum within the six operations is $\lambda \mathbf{p} \cdot \mathbf{p}$ with λ a constant, and the square W of Eq. (??) is assembled from three things derived separately, the 48 forcing the form's isotropy (every quadratic invariant a multiple of $\mathbf{p} \cdot \mathbf{p}$), the 3 the chain's $1/c^2 = 3$ and $E_0'^2$ the band's bottom; "generic" in the three tests of every rule means exactly this. The definitions and this road, with their code and tests, are the group-structure page of the tree [28]; this paper writes no second definition.

The transition from ordinary physics to the algebra, dated. The twenty-four dated steps, each a thing of physics becoming an object of the law with the reading that showed it, are [38] (docs/designs/algebra_transition/HISTORY.md on main at 412f5c61), their dates once: 2026-09-17 the shared resource deleted and the amplitude a phase with a content; 09-18 the Born table from N_ϕ and a Node its six Ports; 09-19 the quantum an integer row, the click the one border, the push one bilinear form, $E = hf$ a release's cost; 09-20 the readings by kind, the masses the initialisation, every count an accumulator, the fan a width of the law; 09-21 the six verbs, the record in $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, the lattice the translation group, the Lorentz factor refused as a seventh verb; 09-22 the detector's clock, the click theorem and the framing.

The platform: how a formula is arrived at, by the algebra and by the simulator. Every formula of this paper is arrived at on one platform, by two roads that meet at the detector. The algebraic road has three steps. The Inside step: the six verbs on the record's attributes, the family's declared integers naming which attributes the step reads (Section 6.1; the families table [52]). The conversion: an emitter puts a record on the board and a detector ends it at a click, so that Outside there are counts and Nodes only, the detector's own count the age wall's member at coefficient 1 (P11). The Outside form: a ratio or a difference of those counts, the image of the Inside step, with its rung (exact, in the fan's limit, in the shell mean) and its order in the grain; a known form, Einstein's or Young's, stands on the comparison side only. The simulator's road runs the same three steps: a world file declares the families, the emitters and the detectors, and the reading is pinned before the run. Every reading of the simulator's road is DETECTOR in its count (each series with its detector in the conversion table [52]), the self-creations' ticks of series S GAMEBOARD; where a count is taken at an open-face detector without a body (the redshift rows 3, 4b and 12) it is read at head in the detector's own clock, its count of self-creations under the age wall ($r = 1.0000$, no crowd at the detector; [49], the re-run of 2026-09-22), equal to the host's interval to the integer, DETECTOR; the registered run's time base was the host's interval [6, record 768], superseded; a two-way light clock at the detector, not read, reads the same count in no crowd (series X); and a number of the board's state is a diagnostic, never a reading. A formula enters the paper with both roads: the algebra's chain with its inputs and its word (the roads table [52]), the simulator's reading with its series and kind (the conversion table [52] and the $S(N_\phi)$ figure of the engine before 2026-09-23, history [52]), and nature on the comparison side (the full record [51, 52]). The platform is the paper's offer beyond its list: to arrive at a further formula, declare the family and the attributes it reads, write the Inside step on them, name the detector and what it counts, take the ratio Outside, pin it before the run, and read it by kind; the three chains below do this in full.

The road to each, and the check that nothing was derived from them. The three roads are the one chain of Section 4, each from the Inside step, the six verbs and the hypotheses named where they enter; the road to each formula and the check that no formula of nature was put in are in the records file [52].

8 The comparison with nature, and the failures by cause

Everything compared here is a detector's count or a ratio of counts (Section 4); the readings are compared by kind, and each failure is named with its cause.

8.1 The simulator's checks and the comparison with nature

Three things are kept apart: the gates on the code; what is shown Inside by algebra under named assumptions, which needs no experiment; and what is measured, a run Inside read Outside, compared

with known experiments.

The formulas after a detector. Every number read after a detector is a count of clicks or a click's tick at a declared detector, registered with its source fingerprint (Appendix B; [4, 7]), its expectation written before the run; a number of the GameBoard's state appears nowhere. Seven formulas shown Inside (the pace, the click's power, Mach-Zehnder, the marginals, GHZ, Malus, light beside a mass) were confirmed by their runs on the engine before 2026-09-23, history (Appendix B); the readings that no formula gives are the conversion table's [52] (the clock's field at two Nodes, the equivalence after a detector, the covariant readings) and the $S(N_\phi)$ figure of the engine before 2026-09-23, history [52]'s (S at the same seven grains).

The method. An experiment is a world file: the GameBoard's extents, the blocks placed on it as the bodies of the experiment, the lamps and the detectors, every one a declared object of Section 3; a result is clicks, a count between clicks on a detector's own record or a ratio of such counts; the pin, the number the algebra gives, is written before the run and no pin moves after a reading. Every row of the paper's list passes three checks in order [26]: approved in the algebra, its number following from the one law before any run and closed on the closure page [12]; approved on the GameBoard without pins, an exploratory run reading beside the algebra's number within the band declared before the run, labelled exploratory and never a result; approved with pins, the pin written before the run [14, 25], one run on a recorded decision, the clicks read by the reader of record and the reading by kind inside the pin. A row is (K), a known-formula row whose algebraic identity is a textbook formula and whose engine reading is the control of the engine against the algebra, or (P), a prediction row, a number the algebra gives and no known formula gives; a prediction is not a pass. A reading that falls outside its pin is written as a miss, in this table, by kind; every number of the law is stated as matching nature or not, never as how nature is. Table 1 is the paper's list at this version: every pinned reading is a blank, since no row of the list has yet been read on the engine as it stands, and the blanks fill as each run is read and merged; the old comparison of the engine before 2026-09-23, its rows, its readings and its figures, is history in the records file [52], never carried into this table.

The limitations, stated before the comparison. The third status follows from the structure of the law as built and never from the outcome, and a referee's test that every cell of the table must pass is that a row sits in the same class had it agreed. The law as built derives a moving clock's rate, $1/\gamma_m$ at the massive kind's cone (Section 3), and not the Lorentz group: the GameBoard is a medium with a rest frame, Lorentz's symmetry belongs to the limit and to the clicks, no Michelson–Morley or Kennedy–Thorndike row is claimed, and the light clock in motion is a predicted FAIL as declared (row LCm); the beam law's bodies that hop had $r = 1$ at every speed and nothing contracted, no relativistic dynamics on them, history; it has no term of the crowd's history, so no observable that reads the growth of an emitter's stretch with the flight time is predicted; and a declared input is not a prediction, so an observable whose number is a declared give (row 7a), a declared content (row 8c) or a declared coefficient (row 13's γ_{PPN}) is not predicted, whatever the declared world reads. A fan of directions bounded in Manhattan length is not a fan of every direction (its angular density is the octahedron's, the axes 2.8 times denser than the face diagonals); a reading that assumes the isotropic mean holds only where the fan is declared uniform in angle at a grain finer than the reading, or where a click read the crowd; otherwise the law's own number for that world is the integral on the comb (the register's finding under rows 6, 10, 13 and 14). The beam law's weight rule for its rows 13 and 14 (γ_{PPN} , the declared space-curvature coefficient) is in the records file [52], history. The count: the paper's list of Table 1 is closed in the algebra before any run, every pinned reading a blank until its run is read and merged, and none is read yet on the engine as

it stands; the old count of the engine before 2026-09-23 is history [52]; the disagreeing rows are the law’s own results, and the paper claims no rule it has not shown. Two placements, row 7a to not predicted and row 2a to NOT COMPARED, were decided on 2026-09-23 after the kind audit [29]; nothing else moved. A row whose apparatus table is the declared law (the polariser’s and the splitter’s tables) is checked as an implementation of that table, and only a row whose evidence class is DERIVED_BLIND is offered as a prediction; the class column of Table 1 fills from the closure page and not from this paper’s judgement.

Table 1 The paper’s list, closed in the algebra before any run [12, 26]: one line per row with its three checks. Kind: (K) a known formula as the thing compared with, (P) a prediction of the algebra with no known formula. Column 1, the algebra’s number, a COMPUTATION declared before the run, with nature’s number beside it for a (K) row; column 2, the GameBoard without pins, an exploratory reading labelled so, or not yet; column 3, the pinned reading by kind, a blank until its run is read and merged, then DETECTOR (a click, a count between clicks) or the word miss. Sources for nature’s numbers as the register cites them [51]. Class: the evidence class of the row, one of DERIVED_BLIND, APPARATUS_INPUT, CALIBRATED_FROM_TARGET, IMPLEMENTATION_GATE and FALSIFIER_ONLY, blank where the closure page has not fixed it (row R2); only a DERIVED_BLIND row is offered as a prediction. Version: every row’s algebra number is the law of 2026-09-23, the massive record kind with the local detector law (on `main` at a867ab22 [9]); no reading of the earlier runtime, the beam law before 3a9a7109, is carried into this table, and wherever the text cites one it is marked history.

Row	Experiment	Kind	1. The algebra (COMPUTATION)	2. The GameBoard, no pins	3. Pinned (by kind)	Class
1a	Bell’s CHSH sum S of a pair, two polariser tables reading the record’s phase	K	the table’s $\cos^2$ law on a shared record summed at the four settings, $S = 181/64 = 2.828$ at the declared grain $N_\phi = 2048$, exact, 1.05 standard errors from Poh et al.: a comparison at a declared grain, not a test of the grain (the measurement excludes the grains whose $S(N_\phi)$ lies outside its band; the shortfall of a measurement from $2\sqrt{2}$ is its apparatus’s); nature 2.42 ± 0.20 [82] and 2.82759 ± 0.00051 [81], verified [16]	not yet: the phase-reading component and the tables	_____	APPARATUS_INPUT
1b	The CHSH sum under the phase-form window	K	2, a control	not yet	_____	APPARATUS_INPUT

Row	Experiment	Kind	1. The algebra (COMPUTATION)	2. The GameBoard, no pins	3. Pinned (by kind)	Class
1c	The order channel	K	the counts no-signalling exactly (row 1d). The order of outcomes: under the counter wheel $[1, N]$ a FAIL by derivation (the second party's outcome sequence flips with the first party's setting on 1448 of 2048 births at $b = 3N/8$, the flip fraction $E_N = S_N/4$ exactly, open by Bell's theorem for any deterministic wheel [15]); under the declared form (the residues in a seed-set order, one permutation per pair lamp drawn once at load from the seed and repeating every W births, the click stamped without the residue [6, records 1647 and 1648]) PASS IN FORM Outside: no reader without the seed reads the first party's setting from the second party's list (exact for a uniformly random order, the seed-set permutation its integer form), the flips against the residues a GAMEBOARD diagnostic of the declared Inside nonlocality; the finite grain's residue, the second party's sequential agreement rate carrying the first party's switching pattern at order $E_N/(W - 1)$, 3.5×10^{-4} at $N = 2048$, is the model's own prediction (COMPUTATION, [22, item 21])	not yet (the declared form's run)	_____	APPARATUS_INPUT
1d	No-signalling of the marginals	K	0 exactly by the table's symmetry, each party's + fraction $N/2$ of N at every setting, 1024 of 2048 on the declared world	not yet	_____	APPARATUS_INPUT
2a	The two slits, one quantum at a time, the visibility	K	at or above the train's coherence value, the band 0.01 (0.959, 0.958, 0.958 at 12, 16, 24 Links for 32 periods); no nature number, NOT COMPARED	partial: the slits' exploratory page	NOT COMPARED, the placement of the limitations paragraph	APPARATUS_INPUT
2b	The Mach-Zehnder visibility	K	1.00 – 0.02, the dark port 0 of 64 at 128 periods; nature 0.98 [83]	not yet: the pair's two arms and the splitter's table	_____	APPARATUS_INPUT
2c	Born's exponent, the three-opening sum	K	the exponent 2 exactly, the click a quadratic form; the sum 0 to the remainder's grain; nature 0.0064 ± 0.0119 [62]	not yet	_____	DERIVED_BLIND
9	Malus at 45 degrees and at 11.25, 28.125, 33.75 degrees	K	128 of 256 at 45 degrees, 0 crossed, 64 with a third between; the table's counts of 256 at the three settings; nature $\cos^2$	not yet on the engine as it stands	_____	APPARATUS_INPUT
10	The single opening's spread, two halves	K	(a) 0.842 at $w = 2\lambda$, $F = 0.16$, the Rayleigh–Sommerfeld sum on the board; (b) 0.886 in the far field; both COMPUTATION [12, row 10 and its script]	not yet	_____	DERIVED_BLIND
7	The atom's lines at the coupled modes	P	the lines at the modes' own frequencies, 0.0995 and 0.1432 radians per interval	exploratory, GAMEBOARD: the coupled mode 0.09948 within 0.3 percent	_____	DERIVED_BLIND
4a	The muon in flight, the moving clock at $k = 3$	K	0.8116 by the one formula at the exact cone on the declared layer (the pair [800, 809], $s = 14$), 0.8108 at c_m and 0.8132 at c beside as controls; nature's form $1/\gamma$ [60]	at rest: the well's mode period 42.364 against 42.36, no beat; in motion partial, the ramp redeclared	_____	DERIVED_BLIND
4b	The moving lamp's redshift	K	$1 + z = 1.9889$ on the declared world, the free limit 1.9339 beside, 1.9355 with the pair [800, 809]; nature's form $\gamma(1 + \beta)$ [61]	not yet: the redeclared geometry	_____	DERIVED_BLIND

Row	Experiment	Kind	1. The algebra (COMPUTATION)	2. The GameBoard, no pins	3. Pinned (by kind)	Class
4c	The round-trip Doppler off a receding transponder	K	$(1 + \beta_c)/(1 - \beta_c) = 3.732$ at $k = 3$ exactly, the hop's sidebands named beside; nature the two-way form of every radar	not yet	_____	DERIVED.BLIND
5a	The anisotropy of c by direction	K	the phase pace by direction ($k^2/36$, Eq. (11)) within 0.8, 0.4, 0.2 percent of c at 12, 16, 24 Links, falling as the inverse square of the wavelength, GAMEBOARD; a PARAMETER BOUND, not an agreement: nature below 10^{-17} [58], 10^{-18} beside [59], the bound on the free grain [15]	not yet on the engine as it stands	n/a, a bound, no pinned entry	DERIVED.BLIND
R2	The Sagnac ratio on two pushed blocks	K	$\beta = v_c/c = 0.5774$ at $k = 3$ exactly, a count between clicks independent of the clock's factor; nature's fringe shift 0.230 ± 0.005 against the computed 0.236 [23] (the two inputs verified in published excerpts, [16, row 80]), the ratio 0.975 ± 0.021 ours, a CONVERSION; the clicks declared at $W = 256$: 108 at rest, 247 chasing, 72 meeting, each ± 2 ; raw $175/319 = 0.5486$, 0.5774 after each direction's own rise, the identity (DECLARATION, sections 13 and 15 at 25a7abf4 [14]; no reading yet)	not yet	_____	_____
LC	The light clock: one emitting block A and a mirror	K, P	the chain's closed face at $x = 672$ sixty Links from A's face; $N_0 = 213 \pm 1$ in A's own clicks at $W = 64$ on its returned record at the receiving set bound to A by name, the free Node at $x = 612$ (the set's first rung at age 213 by the derivation under the click line and the receiver by name, the geometry alone, the declarations' section 10, item 9, merged at ecebf895 [14]; the earlier form's 218 ± 2 , derived for the set at A's twelve cells, no longer in the world, kept as history), a pin declared before any pin run, $2L/c = 207.85$ beside as the (K) form, the rung's rise of about ten intervals the click's own; in motion (P) as declared, $\gamma^2/\gamma_m = 1.2224$ along and $\gamma/\gamma_m = 0.9981$ across (item 7), the light-held pair 1.0136 (item 8); "two bodies" and "two ring-ups" history	not yet	_____	DERIVED.BLIND
LCm	The light clock in motion, Michelson–Morley's null	P	the pushed pair at a declared separation: $\gamma^2/\gamma_m = 1.2224$ along and $\gamma/\gamma_m = 0.9981$ across at $k = 3$ by the band, COMPUTATION [13]; Einstein's 1 and 1 need the separation to contract, so the along arm is a predicted FAIL as declared (a rigid arm's anisotropy $\beta^2/2$, 5×10^{-9} at the Earth's orbital speed, against the cavity bounds at 10^{-17} [58, 59]) and the across arm is met in form, the residue $c_{\text{eff}}/c = 0.99627$ beside; nature's null [58] the falsifier; a light-held pair on the band (the declared world $\lambda_0 = 32.1$ Links at $k = 3$) reads $L_{\text{along}}/L_0 = 0.8127$ against $1/\gamma = 0.8165$ and the along ratio 1.0136 inside the declared band 1 ± 0.03 , COMPUTATION [18], the excess 1.36 percent 42 times $(a/\lambda)^2\beta^2 = 3.24 \times 10^{-4}$ on the declared world (COMPUTATION), tending to $(a/\lambda)^2\beta^2$ only in the limit, and the pair's settling at the nodes the named open step	not yet	_____	DERIVED.BLIND (P)

Row	Experiment	Kind	1. The algebra (COMPUTATION)	2. The GameBoard, no pins	3. Pinned (by kind)	Class
v	The index of a medium block at rest	K	$n^2 = 1 + Gg/(\omega_0^2 - \omega^2)$ in closed form, a control of the coupling; the cavity's mode sum beside it, the 4 percent between them ours, the cavity's mode sum against the closed form, open in size	exploratory at rest: $n - 1$ read at 0.87 of the closed form at all three couplings, a GAMEBOARD probe [19, the index block's row]	_____	DERIVED_BLIND
v-m	The index in motion	P	the scheme's own number on the declared geometry; Fizeau's form the declared non-match	partial, GAMEBOARD: head-on 0.498 beside	_____	DERIVED_BLIND
ii	The moving block's clock on 64^3 , two worlds	P	0.7814 and 0.8032 (clock ratios, dimensionless), the controls beside	exploratory: 0.7833 and 0.8055 by the peak, the three gammas indistinguishable at this board	_____	DERIVED_BLIND
M1	Matter waves through two openings	K	the side lobe's count centroid at $y = 64 + 27.79$ for $\lambda_{dB} = 12$ Links, the two-source sum with the band's k , the band one Node; nature's form de Broglie's	not yet	_____	DERIVED_BLIND
M2	The energy of a moving mass, $E = mc^2$ its rest limit	K	the first click 169 ± 2 intervals after the lamp's birth at the declared rung, the group transit 167.1 beside as the form; $m = k/v_g = 1.0417$ the band's number in the form of a moving mass	not yet	_____	DERIVED_BLIND
A	The two-pace bound on the one scale	P	the deficit $\omega_0^2/4$ of c_m below c ; with the electron's c coefficient 2×10^{-14} [20] one interval is at most 3.6×10^{-28} s, with the one-direction Crab bound at most 6.3×10^{-31} s; a bound, no run	n/a	n/a, a bound	DERIVED_BLIND
A2	Light against itself, the quadratic dispersion by wavelength and direction	P	the coefficient $3 \sum_i n_i^4 - 1$ in $[0, 2]$ on the sky, Eq. (11), a prediction of the form; the bound on the one scale from the burst bound on quadratic photon dispersion, about 2×10^{-35} s, the published number verified against its source [16], the factor $3/2$ the paper's own reading, the sensitivity $\sqrt{2/3}$ beside [27]	n/a	n/a, a form and a bound	DERIVED_BLIND (the form); FALSI-FIER_ONLY (the bound, its source verified)
B	The bound clock's second term	P	$\epsilon(\gamma_m^2 - 1)/2$ below $1/\gamma_m$, printed on every massive world; a bound, no run	n/a	n/a, a bound	DERIVED_BLIND

The derivations, final. Four numbers of the list are the algebra's identities and stand whatever the runs read: Born's exponent 2 exactly, the click a quadratic form (row 2c); Tsirelson's value $2\sqrt{2}$ as the limit of the finite-grain Bell value, the local bound beside it (row 1a); $c^2 = 1/3$, the pace of light as the norm of the flight operator, one Link per interval on every line and $1/\sqrt{3}$ in the Euclidean mean (row 5a); and the rest energy of a massive kind, $h\omega_0$ with $\omega_0 = mc^2$ in the law's units at the massive cone, de Broglie's internal clock as the thing compared with (row M2).

Under the hypothesis, after the pins, on a recorded decision. Three rows enter only under the pair-field hypothesis, a third record kind sourced by every block's content and read by every Node as its own pair, outside the law under its own identity and never in the list's place: the clock in a field at r and $2r$, the ratio 2.00 (row 12, Newton's potential from the lattice Green's function); the delay of a light pulse through a medium block sitting in the field, Shapiro's form with the same $1 + \gamma$, one declared input (in place of row 13's bending); and Newton's fall, a prediction of the form and not a pinned row (in place of row 14's orbit).

The rows that left the list. Under the law as it stands the following rows of the register are not predicted, each with its missing piece named on the schedule [25], and their earlier readings are history in the records file [52]: the far lamp (rows 3 and 11a to 11c, a crowd's field from the rule), the nuclear bindings (7a and 7b, a binding energy from the rule), the decays and the neutrino (8a to 8c, a lifetime and a mass ratio from the rule), the two arms in motion (5b, kept as the prediction page of the light-bound arms), complementarity and aberration (R5 and R6, their declarations), Bohr's ladder (6, replaced by the atom's lines), the bending (13, replaced by the delay under the hypothesis) and the period ratio (14, replaced by the fall under the hypothesis). No row is removed from the register; a row not predicted says so.

The prediction rows. The (P) rows of Table 1 are the algebra's own numbers with no known formula behind them: the atom's lines at the coupled modes, the index in motion, the moving block's clock on the two 64^3 worlds, the ring-ups of the light clock, the two-pace bound, light's anisotropic dispersion on the sky and the bound clock's second term; each has its falsifier written before any run [9, 8.9], and a prediction is not a pass. The rows read on the engine before 2026-09-23, with their agreements and their disagreements, are history [52]; what this version claims is the algebra's numbers before the runs, and the blanks of the table say so.

8.2 Discussion and conclusions

The chain from the click to Lorentz's factors, Einstein's step and Newton's form is Section 4; the two roads and their check are Section 7; what is proved opens Section 5.

What the law names and has not computed. Each item below is a computation the law's rules define, on a declared world with a declared click, that this paper has not made: the register holds no reading for it, nothing is claimed for it, and its verdict is NOT YET, NOT COMPARED or absent until a run is made and read by kind. (i) The lines of the atom: not a run not yet made but a rule the law lacks, no rule of the six quantising a body's loop and one click reaching only the loops with $2i^2 \geq j^2$ (the full record [51, 52], row 6; [46]); Bohr's 27/20 waits on that rule, under its own identity; the ladder itself, $a_j/a_i = (j/i)^2$ and $E_j \propto 1/j^2$ with 27/20, is conjectured from the algebra, not computed (the atom's item in the records file [52]). (ii) Third-order interference, Sorkin's κ_S on a three-opening world (row 2c; no such world is registered). (iii) The single opening's spread against its pin (row 10), and the rows pinned without their run (4a, 5b, 11). (iv) The quarks and the nucleon: the quark families are in the catalog with their rational charges and their content per record (the families table [52], u), and no detector reading of a quarks world is registered; the nucleon as three measured events bound by the strong column, its read mass against the sum of its parts, is the reading that would test them [6, record 248]; colour, a $\mathbb{Z}_3$ label that passes the three tests and does nothing beside a colour force between rows that fails the vector and the local test, is designed under its own identity and not built [6, record 270]. (v) The unseen content: on the law's own rules the field at a galaxy's edge is Newton's in the shell mean and the detector adds nothing there; the one route to a flat rotation curve without unseen content, a short periodic dimension ([31], entry 6), fails nature's Tully-Fisher slope on a host map ([48], no run); what the law admits is unseen content as an input family, as nature's model does; no rotation curve is read after a detector and no disc under a declared unseen family is run. (vi) The size of the crowd's stretch, $g_1 = 0.36$, that the supernova diagram needs, from a declared content over a flight (an input in the dark energy item of the list above; the crowd's own accumulation gives the wrong sign), and the periodic GameBoard against the open one on the measured numbers, declared and not decided [6, record 270]. (vii) The detector's constants c_j of the click's harmonics, not reached by the algebra (Section 5.3). (viii) The inverse square in space, pending its run (row 14); the bending's

full deflection, its coefficient $2(1 + \gamma)$ on the one constant conjectured from the algebra; the ring's rule read inside its pins and the constant 4 not closing within the grain (row 13). Apart from these stand the numbers for which the law has no formula and this paper offers none: the hierarchy ρ_p , the constant $h = h_q N_\phi$, the families' content per unit and the grain N_ϕ , inputs by the dictionary (the forcing ledger [52]).

Positioning. The click's assignment of the second outcome from the joint weights is a non-local step on the labels, its interval the last arm's (as after Theorem ??), forced by Bell's theorem because the marginals are exact and $S > 2$ [53, 54]; no-signalling of the counts is Theorem ?? [55], and the order of the outcomes signals under the sequential wheel (Section 5.4; the full record [51, 52], row 1c, FAIL BY DERIVATION, the order channel computed before any run [15]). The click's finite kinematics is Weyl's and Schwinger's on $\mathbb{Z}_{N_\phi}$ [66, 67]; the transport a lattice gas's exact integer dynamics [56, 57], the flight table's transport free of dispersion where the lattice automata of [71, 72, 73] disperse, the record kind's not, its k^2 dispersion the prediction of row A2; the uniform u does the work of quantum equilibrium [74, 75]; the model is ψ -ontic [76, 77]; Theorem ?? puts $S(N_\phi)$ on both sides of Tsirelson's bound [79, 80]. The earlier testbed [1] is history.

Conclusion. One local integer rule on the cubic lattice, six operations required to commute with the group of order 48, is read only through detectors' clicks. Exact on the lattice are the count marginals, the read-out's quadratic form under its axiom and the CHSH sum as a rational of the grain; the moving clock's rate is derived under the massive record kind; Lorentz's factors and Einstein's step are reached only under two named hypotheses. No pinned reading is in the table yet: every row is closed in the algebra before its run, and the pinned runs decide, row by row, what the law matches and where it fails.

Declarations

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Ethics approval. Not applicable.

Consent to participate and consent for publication. Not applicable: no human participants and no animals.

Data availability. Every world file, every registered run's record, the check scripts' outputs and the derivation records cited in this paper are archived at Zenodo [84] (concept DOI 10.5281/zenodo.22738746); the commit of the archived version is named in Appendix B, and every number of the paper has one row in the ledger of that archive (`paper/general_formula/NUMBERS.md`) naming its kind and its source.

Materials availability. Not applicable: no physical materials.

Code availability. The simulator, the check scripts and the figures' scripts are in the same archive [84], under its open licence; the figures are drawn by those scripts from the definitions, none by a generative tool.

Author contributions. Sole author: the guidelines of the project, the model, its rules, the worlds and the experiments, every reading and its verdict, and the decisions recorded in the archived log; the part of the large language model used as a tool is stated in the next declaration.

Use of artificial intelligence. The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under the author’s direction and instructions, for the software of the simulator and its checks, the computations, the mathematical derivations and their verification, the physics of the design pages, and the drafting and editing of the text; the model is not an author and meets no criterion of authorship. Every guideline, postulate, hypothesis, rule, world and decision of the project is the author’s, recorded in the archived log; the author reviewed and edited every statement of this paper and takes full responsibility for its content. The author’s statement: I laid down every postulate and hypothesis of the model myself and brought in nothing external; the forms of Newton, Kepler, Einstein, Lorentz, Bohr and Balmer that this paper cites are the things compared with, not inputs to the model.

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A Technical details

The tables and their rounding. The circle is $\mathbb{Z}_{N_\phi}$ with $4 \mid N_\phi$. The tables are $C_{N_\phi}[p] = \text{round}(256 \cos(2\pi p/N_\phi))$ and $S_{N_\phi}[p] = C_{N_\phi}[p - N_\phi/4]$, so $C_{N_\phi}[p + N_\phi/2] = -C_{N_\phi}[p]$ and $S_{N_\phi}[p + N_\phi/4] = C_{N_\phi}[p]$ exactly; the norm $C_{N_\phi}^2 + S_{N_\phi}^2$ lies in $[65536 - 351, 65536 + 361]$ for every power of two $N_\phi \leq 65536$ [5]. The normalisation by C_K in Eq. (??) makes the click's probabilities sum to one at completion, and the paper claims exact unitarity nowhere.

The symbols. The symbol table, every letter with its kind and its name, is in the paper's records file [52]; every symbol is named at its first use in the text.

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Every registered run's world file, its expectation written before the run, its source fingerprint and its record, the check scripts with their outputs, the figures' runs and the seven confirmations (each a run's clicks against the formula shown Inside, with its numbers) are listed in the paper's records file [52] and in the registers

[4, 7]; the code and the worlds are the archive [84]. The archived version cited at submission is the tag `paper-2026-09-22` on the merge commit of the pull request that carries this cut, one tree whose history holds the trees named above and the log records this paper cites; it replaces the concept DOI of the data availability statement. The software, the computations, the derivations' checks and the text were made with a large language model as a tool under the author's direction and instructions, as the declaration at the end of the paper states; every decision is the author's. Every theorem, identity and number of the paper was re-derived and recomputed independently, one line per statement [17]. The paper's law is the massive record kind with the local detector law, on `main` at `a867ab22` [9]; the beam law's runtime before the generic entry of the bending (the runs' own commits named above, on the first-parent line of `main` up to `3a9a7109`, the last commit before the entry's merge `5b855791`) is history, its flight table and its readings in the records file [52]; the generic entry is a later rule of that runtime (Section 6.1, P9), and the ring's run (the full record [51, 52], row 13) is this paper's one reading under it. Every record, note, design and register row this text cites is on the tree of this cut (main merged into it at `87c7ec6b`).

The long form of this manuscript, with the derivation record of every formula, is the tree's paper at the commit `c15c1174`; the forty-page cut before it at `b0d1ebf7`.